

Practical 7 - Logistic Regression in Cohort and Case-Control Studies

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `oeso1.csv`, `oeso2.csv`

Introduction

We will use data from a case-control study of oesophageal cancer. Cases are 200 men diagnosed with oesophageal cancer in the Ille-et-Vilaine area of France between January 1972 and April 1974. Controls are 776 men who were sampled from an electoral register. The data are used extensively in the book Breslow, N.E. and Day, N.E. 1980. Statistical Methods in Cancer Research: Volume 1 The Analysis of Case-Control Studies. This book is freely available online [here](#).

The first dataset is called `oeso1.csv` and consists of one row per participant, and four variables as below:

Variable	Description
<code>case</code>	0 = control, 1 = case
<code>age_group</code>	Age group 1: 25–34, 2: 35–44, 3: 45–54, 4: 55–64, 5: 65–74, 6: 75+
<code>tobacco_group</code>	Tobacco intake (g/day) 0: None, 1: 1–9, 2: 10–19, 3: 20–29, 4: 30+
<code>alcohol_grp</code>	Alcohol intake (g/day) 0: 0–39, 1: 40–79, 2: 80–119, 3: 120+

Aims

We will examine the association between oesophageal cancer as the outcome and a dichotomous smoking status variable (smoker or non-smoker) as an exposure, and attempt to answer the following questions:

1. Does smoking increase the risk of oesophageal cancer?
2. Does alcohol intake or age confound this relationship?
3. What is the best way to adjust for these potential confounders: as categorical or continuous variables?

We will also swap the exposure and outcome variables to explore what happens to the estimate of the odds ratio (and its confidence interval).

Analysis

Question 1

Load `oeso1.csv` using `read.csv`. Then use `factor` to convert `age_group`, `tobacco_group` and `alcohol_grp` into factors with the correct labels.

Create a new variable, `tobacco_2` that takes value 0 if they are in `tobacco_group` 0 and 1 otherwise.

Then explore the associations between:

- a) binary tobacco status and alcohol group
- b) binary tobacco status and age
- c) alcohol and age

Discuss: Are there any associations which may affect your analysis?

SOLUTIONS

```
oeso1 <- read.csv("oeso1.csv")
oeso1$age_group <- factor(oeso1$age_group,
                          labels = c("25-34", "35-44",
                                     "45-54", "55-64",
                                     "65-74", "75+"))
oeso1$tobacco_group <- factor(oeso1$tobacco_group,
```

```

labels = c("None", "1-9", "10-19", "20-29", "30+")
oeso1$alcohol_grp <- factor(oeso1$alcohol_grp,
labels = c("0-39", "40-79", "80-119", "120+"))

oeso1$tobacco_2 <- 0
oeso1$tobacco_2[oeso1$tobacco_group!="None"] <- 1
oeso1$tobacco_2 <- factor(oeso1$tobacco_2,
labels= c("No tobacco", "Some tobacco"))

```

Now we examine the associations.

```
prop.table(table(oeso1$tobacco_2, oeso1$alcohol_grp),1)
```

	0-39	40-79	80-119	120+
No tobacco	0.56060606	0.31439394	0.10227273	0.02272727
Some tobacco	0.37359551	0.38342697	0.15730337	0.08567416

```
prop.table(table(oeso1$age_group, oeso1$tobacco_2),1)
```

	No tobacco	Some tobacco
25-34	0.4273504	0.5726496
35-44	0.3467337	0.6532663
45-54	0.2629108	0.7370892
55-64	0.2066116	0.7933884
65-74	0.1614907	0.8385093
75+	0.2954545	0.7045455

```
prop.table(table(oeso1$age_group, oeso1$alcohol_grp),1)
```

	0-39	40-79	80-119	120+
25-34	0.52991453	0.38461538	0.04273504	0.04273504
35-44	0.43718593	0.41206030	0.10050251	0.05025126
45-54	0.36619718	0.38028169	0.18309859	0.07042254
55-64	0.36776860	0.34297521	0.18181818	0.10743802
65-74	0.44099379	0.32919255	0.18012422	0.04968944
75+	0.61363636	0.27272727	0.04545455	0.06818182

There appears to be an association between tobacco intake and alcohol intake, with people with high tobacco intake tending to have higher alcohol intake than people using no or small amounts of tobacco. Those using no tobacco or in the highest tobacco intake group tend to be younger than those in the middle categories of tobacco intake. We may expect alcohol intake and age as well as tobacco intake to be associated with the outcome, oesophageal cancer. These preliminary investigations suggest that tobacco is associated with both alcohol and age and it is therefore sensible to consider these variables as potential confounders in our analysis.

Question 2

- Use a two-by-two table for the binary tobacco variable and the case-control status to estimate the odds ratio. Use Woolf's method to obtain a 95% confidence interval for this estimate.
- Interpret your results.

SOLUTIONS

```
table(oeso1$tobacco_2, oeso1$case)
```

	0	1
No tobacco	255	9
Some tobacco	521	191

```
or = (255*191)/(9*521)      # Odds ratio
var = 1/255+1/191+1/9+1/521 # Var of estimate (that is SE^2)
se = sqrt(var)

# Use the variance of the estimate to calculate Woolf's 95% CI
or * exp(-1.96*se) # Lower
```

```
[1] 5.23537
```

```
or * exp(1.96*se) # Upper
```

```
[1] 20.60816
```

The estimated odds ratio is $(255 \times 191) / (9 \times 521) = 10.39$ and the estimated variance of the log OR estimate is $1/255 + 1/191 + 1/9 + 1/521 = 0.122$ (this is Woolf's estimate of the variance). A 95% CI for the odds ratio is therefore (5.24, 20.61). There is a strong and highly statistically significant association between use of tobacco and oesophageal cancer.

Question 3

Write down the algebraic form of the following logistic regression models, and then use the `glm` function to fit the models:

- a) with oesophageal cancer as the outcome and tobacco intake as the explanatory variable
- b) with tobacco intake as the outcome and oesophageal cancer as the explanatory variable

Discuss: Compare and contrast the results from the two analyses.

SOLUTIONS

Let $Y = 1$ denote a case and $Y = 0$ denote a control, and let $X = 1$ denote a smoker and $X = 0$ a non-smoker. Using the notation from the lecture notes, the logistic regression model for (a) is

$$\Pr(X = 1|Y = y) = \frac{e^{\lambda^* + \beta y}}{1 + e^{\lambda^* + \beta y}}$$

The logistic regression model for (b) is

$$\Pr(Y = 1|X = x) = \frac{e^{\alpha^* + \beta x}}{1 + e^{\alpha^* + \beta x}}$$

The results from fitting the two models in R are:

```
logistic_tobac_case <- glm(tobacco_2~case, data=oeso1,
                           family=binomial)
summary(logistic_tobac_case)
```

Call:

```
glm(formula = tobacco_2 ~ case, family = binomial, data = oeso1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.71449	0.07643	9.349	< 2e-16 ***

```

case          2.34056      0.34951      6.697 2.13e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 1139.5  on 975  degrees of freedom
Residual deviance: 1056.1  on 974  degrees of freedom
AIC: 1060.1

```

Number of Fisher Scoring iterations: 5

```

logistic_case_tobac <- glm(case~tobacco_2, data=oeso1,
                           family=binomial)
summary(logistic_case_tobac)

```

Call:

```
glm(formula = case ~ tobacco_2, family = binomial, data = oeso1)
```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -3.3440     0.3392  -9.860  < 2e-16 ***
tobacco_2Some tobacco  2.3406     0.3496   6.696 2.14e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 989.95  on 975  degrees of freedom
Residual deviance: 906.59  on 974  degrees of freedom
AIC: 910.59

```

Number of Fisher Scoring iterations: 6

The log odds ratio estimates from models (a) and (b) are identical, as we expect. Also as we expect, the intercept terms are different. In model (a) the intercept λ_0^* is the log odds of being a smoker among the controls. So 0.714 is the estimated log odds of being a smoker among the controls. The intercept in model (b) does not have a sensible interpretation - see the lecture notes. The form of intercept α^* is in the lecture notes.

Question 4

We will now control for alcohol intake. Write R code to implement the calculations for the Mantel-Haenszel method, and find the conditional (on alcohol intake) odds ratio estimate:

- a) with oesophageal cancer as the outcome and tobacco intake as the explanatory variable
- b) with tobacco intake as the outcome and oesophageal cancer as the explanatory variable

SOLUTIONS

We can use `table` with three arguments to get the series of 2x2 tables across levels of `alcohol_grp`, if we put `alcohol_grp` as the third variable:

```
table(oeso1$case, oeso1$tobacco_2, oeso1$alcohol_grp)
```

```
, , = 0-39
```

	No tobacco	Some tobacco
0	148	237
1	0	29

```
, , = 40-79
```

	No tobacco	Some tobacco
0	80	201
1	3	72

```
, , = 80-119
```

	No tobacco	Some tobacco
0	22	66
1	5	46

```
, , = 120+
```

	No tobacco	Some tobacco
0	5	17

1

1

44

```
#we do the calculations 'by hand' here
or_mh_num <- (148*29)/(148+237+29) + (80*72)/(80+201+3+72) +
  (22*46)/(22+66+5+46) + (5*44)/(5+17+1+44)
or_mh_denom <- (237*0)/(148+237+29) + (201*3)/(80+201+3+72) +
  (66*5)/(22+66+5+46) + (17*1)/(5+17+1+44)
or_mh_num/or_mh_denom
```

```
[1] 8.587244
```

We do not do the second calculation - we can see from the formula for Mantel-Haenszel in the notes that if we swap the role of outcome and explanatory variable (exposure) the value of the estimate will not change.

The adjusted OR is 8.59, as compared with the unadjusted OR of 10.39. That is, the OR for being a case in smokers compared with non-smokers conditional on alcohol intake is slightly attenuated (closer to 1).

Question 5

Repeat Q4 a) and b) using `glm`.

Discuss: Compare and contrast the results from the two models, and the two Mantel-Haenszel calculations you performed. What do the intercept parameters represent in each logistic model?

SOLUTIONS

```
alcAdj_case_tobac <- glm(case~tobacco_2+alcohol_grp,
  data=oeso1,
  family=binomial)
summary(alcAdj_case_tobac)
```

Call:

```
glm(formula = case ~ tobacco_2 + alcohol_grp, family = binomial,
  data = oeso1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.3059	0.3854	-11.173	< 2e-16 ***
tobacco_2Some tobacco	2.1265	0.3588	5.926	3.10e-09 ***
alcohol_grp40-79	1.1456	0.2366	4.841	1.29e-06 ***
alcohol_grp80-119	1.9182	0.2682	7.154	8.46e-13 ***
alcohol_grp120+	3.0994	0.3350	9.251	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 989.95 on 975 degrees of freedom
Residual deviance: 786.56 on 971 degrees of freedom
AIC: 796.56

Number of Fisher Scoring iterations: 6

```
exp(coef(alcAdj_case_tobac))
```

(Intercept)	tobacco_2Some tobacco	alcohol_grp40-79
0.01348878	8.38560125	3.14437933
alcohol_grp80-119	alcohol_grp120+	
6.80884332	22.18507867	

```
alcAdj_tobac_case <- glm(tobacco_2~case+alcohol_grp,  
                          data=oeso1,  
                          family=binomial)  
summary(alcAdj_case_tobac)
```

Call:

```
glm(formula = case ~ tobacco_2 + alcohol_grp, family = binomial,  
     data = oeso1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.3059	0.3854	-11.173	< 2e-16 ***
tobacco_2Some tobacco	2.1265	0.3588	5.926	3.10e-09 ***
alcohol_grp40-79	1.1456	0.2366	4.841	1.29e-06 ***
alcohol_grp80-119	1.9182	0.2682	7.154	8.46e-13 ***

```
alcohol_grp120+      3.0994      0.3350      9.251 < 2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 989.95  on 975  degrees of freedom  
Residual deviance: 786.56  on 971  degrees of freedom  
AIC: 796.56
```

```
Number of Fisher Scoring iterations: 6
```

People who use tobacco have an odds of oesophageal cancer which is over 8 times that in non-users of tobacco, after conditioning on alcohol intake. Both the intercepts and the coefficients for the alcohol categories are different in models (a) and (b). In model (b) the intercept is the log odds of being a smoker in controls who are in the lowest alcohol intake group. In model (a) the intercept does not have a sensible interpretation.

From Model (a) we can see that increasing alcohol intake is independently associated with higher odds of oesophageal cancer after conditioning on smoking status. People consuming 120+ grams of alcohol per day had an odds of oesophageal cancer which is 22 times greater than that of those in the lowest alcohol intake group (0-39 grams per day), after conditioning on tobacco intake.

The coefficients for alcohol intake in model (b) relate to their association with smoking status, after conditioning on case/control status. People consuming 120+ grams of alcohol per day had an odds of being a smoker which is 2.3 times greater than that of those in the lowest alcohol intake group, after conditioning on case/control status. This is not likely to be an interpretation that is of interest to us.

Question 6

We will now also condition on age. Add the age group variable into the model with case-control status as the outcome and interpret the results.

Discuss: What impact does adding age to the model have on the odds ratio for tobacco status? What are two possible reasons for any impact that you see?

SOLUTIONS

```
alcAgeAdj_case_tobac <- glm(case~tobacco_2+alcohol_grp+age_group,
                             data=oeso1,
                             family=binomial)
summary(alcAgeAdj_case_tobac)
```

Call:

```
glm(formula = case ~ tobacco_2 + alcohol_grp + age_group, family = binomial,
     data = oeso1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.6714	1.1073	-6.928	4.27e-12 ***
tobacco_2Some tobacco	1.9663	0.3741	5.256	1.47e-07 ***
alcohol_grp40-79	1.3498	0.2498	5.403	6.56e-08 ***
alcohol_grp80-119	1.8679	0.2840	6.577	4.81e-11 ***
alcohol_grp120+	3.5125	0.3831	9.168	< 2e-16 ***
age_group35-44	1.5806	1.0876	1.453	0.146132
age_group45-54	3.3632	1.0468	3.213	0.001314 **
age_group55-64	3.8336	1.0419	3.680	0.000234 ***
age_group65-74	4.1962	1.0488	4.001	6.31e-05 ***
age_group75+	4.3755	1.1052	3.959	7.52e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 989.95 on 975 degrees of freedom
 Residual deviance: 686.55 on 966 degrees of freedom
 AIC: 706.55

Number of Fisher Scoring iterations: 7

```
exp(coef(alcAgeAdj_case_tobac))
```

(Intercept)	tobacco_2Some tobacco	alcohol_grp40-79
4.659617e-04	7.144524e+00	3.856561e+00
alcohol_grp80-119	alcohol_grp120+	age_group35-44
6.474554e+00	3.353352e+01	4.857925e+00
age_group45-54	age_group55-64	age_group65-74

2.888156e+01
age_group75+
7.947950e+01

4.622982e+01

6.643176e+01

Adding age group into the model has reduced the OR for tobacco intake to about 7. The change in the OR compare to the model with adjustment for alcohol intake only could be due to a combination of confounding by age and non-collapsibility. The investigations in question 1, and subject-matter knowledge suggest that age is expected to be a confounder. The inclusion of age has not had much of an effect on the odds ratios for alcohol intake. Increasing age is strongly associated with oesophageal cancer risk given tobacco use and alcohol intake.

Question 7

Age and alcohol intake are in fact available as continuous variables. These can be found in the `oesophageal_data-2.csv` dataset. Load the data, and fit a logistic model for oesophageal cancer with three explanatory variables: binary tobacco intake, continuous alcohol intake and continuous age.

Discuss: Compare the results from this model to your results from Q6. What are the advantages and disadvantages of the two approaches?

SOLUTIONS

```
oeso2 <- read.csv("oeso2.csv")
#we'll skip coding the categorical variables into factors here
oeso2$tobacco_2 <- 0
oeso2$tobacco_2[oeso2$tobacco_group!=0] <- 1
oeso2$tobacco_2 <- factor(oeso2$tobacco_2,
                          labels= c("No tobacco", "Some tobacco"))

oeso2_mod <- glm(case~tobacco_2+alcohol+age, data=oeso2,
                 family=binomial)
summary(oeso2_mod)
```

Call:

```
glm(formula = case ~ tobacco_2 + alcohol + age, family = binomial,
     data = oeso2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.230225	0.651144	-12.640	< 2e-16 ***
tobacco_2Some tobacco	1.927582	0.366357	5.261	1.43e-07 ***
alcohol	0.025983	0.002542	10.220	< 2e-16 ***
age	0.064660	0.008172	7.912	2.53e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 989.95 on 975 degrees of freedom
Residual deviance: 702.99 on 972 degrees of freedom
AIC: 710.99

Number of Fisher Scoring iterations: 6

```
exp(coef(oeso2_mod))
```

(Intercept)	tobacco_2Some tobacco	alcohol
0.0002664763	6.8728706141	1.0263239171
age		
1.0667963135		

There is still a strong association between smoking and oesophageal cancer conditional on alcohol intake and age, though the odds ratio estimate has changed a little from that found above. When including age and alcohol intake as continuous variables, we need to choose a suitable form in which they will be entered into the model. Here we assumed that age and alcohol act linearly on the log odds of being a case. However, a non-linear association may exist in reality. In general, when using continuous variables it is a good idea to investigate whether non-linear terms are needed, e.g. by including quadratic and cubic terms. Alternatively, age and alcohol intake could be included using flexible non-linear functions, e.g. using splines (see later session on non-linear effects). The most compelling reason to include age and alcohol intake as continuous variables is that by doing so we use the full information on these variables. By categorising variables we throw away information.

Question 8

Present your results in a table or tables suitable for use in a paper or report.

SOLUTIONS

You can create a table in e.g. Word by hand, or if you want one programmatically for use in an RMarkdown or Quarto report, you could use:

```
resTable <- array(0, dim=c(3,3))
colnames(resTable) <- c("Model", "Odds ratio", "95% CI")

#create helper function that extracts OR and 95% CI from a model
#it assumes the effect of interest is 2nd coefficient in model
tobacEffect <- function(logitModel) {
  estOR <- coef(logitModel)[2]
  logORse <- sqrt(vcov(logitModel)[2,2])
  ciLower <- estOR-1.96*logORse
  ciUpper <- estOR+1.96*logORse
  #return exponentiated values in a string, rounded to 2 decimal places
  c(round(exp(estOR),2),
    paste0("(", round(exp(ciLower),2), ", ", round(exp(ciUpper),2), ")"))
}

resTable[1,1] <- "Unadjusted"
resTable[1,c(2,3)] <- tobacEffect(logistic_case_tobac)

resTable[2,1] <- "Conditional on alcohol and age (as categorical variables)"
resTable[2,c(2,3)] <- tobacEffect(alcAgeAdj_case_tobac)

resTable[3,1] <- "Conditional on alcohol and age (as continuous variables)"
resTable[3,c(2,3)] <- tobacEffect(oeso2_mod)

knitr::kable(resTable,
              caption="Odds ratio (OR) estimates (95% confidence intervals
                        (CI)) for the association between smoking status and oesophageal
                        cancer from unadjusted and conditional/adjusted logistic
                        regression models.")
```

Table 2: Odds ratio (OR) estimates (95% confidence intervals (CI)) for the association between smoking status and oesophageal cancer from unadjusted and conditional/adjusted logistic regression models.

Model	Odds ratio	95% CI
Unadjusted	10.39	(5.24, 20.61)

Model	Odds ratio	95% CI
Conditional on alcohol and age (as categorical variables)	7.14	(3.43, 14.87)
Conditional on alcohol and age (as continuous variables)	6.87	(3.35, 14.09)